PROJECT TITLE:
SUPERSTORE MARKETING CAMPAIGN ANALYSIS

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Introduction:

Initiative: Every year-end, the Superstore chain launches a major promotional campaign inviting existing customers to purchase a Gold membership offering a 20% discount on all purchases.

Dataset: The dataset contains 2,240 customers and 22 features, including the demographics, channel preferences, spending behaviour, purchase recency, complain and response.

Purpose: Use data to understand what influences membership acceptance, identify profitable customer segments, and improve future marketing campaigns.

Objectives:

1. **Customer Segmentation**: Divide customers based on demographics and spending behaviour.
2. **Behavioral Analysis**: Identify key drivers such as recency, product category spend, and channel usage.
3. **Predict membership acceptance**: Build classification model such as logistic regression to predict membership acceptance.
4. **Recommend Strategies**: Suggest ways to increase membership uptake, and improve marketing campaigns.

Issues and Challenges:

- ❖ **Missing Values:** Income column contains missing values that need to be filled or removed.
- ❖ **Class Imbalance:** Low response rate creates challenges for predictive modelling accuracy.
- ❖ **Fragmented Behaviour:** Customers use different channels, making it difficult to identify a consistent buying pattern.
- ❖ **Ambiguous Categories:** Some categories in Education and Marital Status were unclear or incorrectly labeled**.**

Key Performance Indicators (KPI):

KPI	Value	Purpose
Response Rate	14.91%	Measure campaign effectiveness
Total Spend	1.36M	Measure total amount spend by customers
Average Spend	605.80	Helps identify customer value

Figure 1: KPIs used in Dashboard

Data Quality Assessment:

I created a data quality summary table in Excel to identify data issues, suggested cleaning action, and measure the count and percentage of affected rows.

Column Name	Issue	Suggested Action	Count	% of Affected rows
Income	Null Values	Impute with Median	24	1.07%
Income	Outlier	Imputation of Median	1	0.04%
Marital_Status	Invalid categories(Absurd,Alone,YOLO)	Merging into "Single"	7	0.31%
Education	Ambiguous category name(2n cycle)	Merge into "Basic"	203	9.06%
Dt_Customer	Format inconsistency	Standardize format	1324	59.11%
Year_Birth	Unrealistic ages(>100 years)	Impute with Median or remove	2	0.09%
ID	Invalid Id value(0)	Replace with new unique ID after validation	1	0.04%

Figure 2: Data quality summary table in Excel

Data Extraction & Validation:

Initial processing was performed in Excel, where date formats were standardized and blank income values were temporarily replaced with 0 to ensure smooth data import into SQL.

The dataset was then imported into SQL by creating a structured table.

Data Validation was carried out by verifying total row counts, resetting income values to NULL where applicable, creating Views for structured analysis, and performing Sanity checks using SQL queries to ensure data consistency and accuracy.

```sql
-- Total row count
Select Count(*) As Total_Rows
from superstore_campaign;
```

Result Grid | Filter Rows:

Total_Rows
2240

Figure 3: Counting total rows

```sql
-- Distinct Counts for fields
Select
        Count(Distinct ID) As Dist_IDs,
        Count(Distinct Education) As Dist_Education,
        Count(Distinct Marital_Status) As Dist_Marital_Status,
        Count(Distinct Response) As Dist_Response,
        Count(Distinct Complain) As Dist_Complain
From superstore_campaign;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Dist_IDs	Dist_Education	Dist_Marital_Status	Dist_Response	Dist_Complain
2240	5	8	2	2

Figure 4: Checking distinct counts for categorical columns

```sql
Create View Customer_info As
Select
    ID,
    Year_Birth,
    (Year(curdate()) - Year_Birth) As Age,
    Education,
    Marital_Status,
    Income,
    Kidhome,
    Teenhome,
    (Kidhome + Teenhome) As Total_Children,
    Dt_Customer,
    TimeStampDiff(Year, Dt_Customer,CurDate()) As Tenure
From superstore_campaign;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

ID	Year_Birth	Age	Education	Marital_Status	Income	Kidhome	Teenhome	Total_Children	Dt_Customer	Tenure
0	1985	41	Graduation	Married	70951	0	0	0	2013-04-05	13
1	1961	65	Graduation	Single	57091	0	0	0	2014-06-15	11

Figure 5: Created View to show Customer demographics

```sql
-- Calculating Min/Max for Numeric Columns
Select
        Min(Id) As Min_Id,
        Max(Id) As Max_Id,
        Min(Year_Birth) As Min_Birth_Year,
        Max(Year_Birth) As Max_Birth_Year,
        Min(Year(CURDATE())- Year_Birth) As Min_Age,
        Max(Year(CURDATE())- Year_Birth) As Max_Age,
        Min(Income) As Min_Income,
        Max(Income) As Max_Income,
        Min(Recency) As Min_Recency,
        Max(Recency) As Max_Recency
From superstore_campaign;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Min_Id	Max_Id	Min_Birth_Year	Max_Birth_Year	Min_Age	Max_Age	Min_Income	Max_Income	Min_Recency	Max_Recency
0	11191	1893	1996	30	133	1730	666666	0	99

Figure 6: Validating min/max values for numeric columns

Data Processing in Python:

The dataset was imported into Python using SQLAlchemy for further analysis and processing.

Data validation was performed to ensure data quality and consistency, including verification of dataset shape, checking data types for all variables, and identifying missing values across columns.

```
df.shape

(2240, 22)

df.describe(include = 'all')
```

Figure 7: Dataset shape verification

```
df.isna().sum()

ID                 0
Year_Birth         0
Education          0
Marital_Status     0
Income            24
Kidhome            0
Teenhome           0
```

Figure 8: checking missing values across dataset columns

Data Cleaning:

Data Cleaning was performed to improve data quality. The following steps are carried out:

- ✓ Missing Values in Income were handled using median imputation.
- ✓ Extreme outlier which is ‘666,666’ was identified and imputed with median.
- ✓ Ambiguous category in Education i.e. “2n Cycle” was merged into “Basic”.
- ✓ Anomalous categories in Marital Status such as “Absurd”, “Alone” and “YOLO” were merged into “Single” category.
- ✓ Inconsistent Date formats were standardized.
- ✓ Invalid ID value (0) were replaced with unique identifiers.
- ✓ Unrealistic Age value were identified (e.g. 1893, 1899, 1900) resulting the ages above 100, and were handled appropriately.

These steps ensured consistency accuracy, and reliability of the dataset for further analysis.

```
df['Income'].isna().sum()
```

```
np.int64(24)
```

```
# Handling missing Income
df['Income']= df['Income'].fillna(df['Income'].median())
```

Figure 9: Handled missing values in Income

```
# Merging '2n Cycle' category into 'Basic'
df['Education'] = df['Education'].replace({'2n Cycle':'Basic'})
```

Figure 10: Merging inconsistent category in Education

Feature Engineering:

New columns were created such as **Age, Tenure, Total Spend, Total Purchase, Category Breadth, Web vs Store Ratio** and **Total Children** to better understand customer behaviour.

```
# Calculating Customer Tenure from Dt_Customer
df['Tenure'] = ((datetime.now() - pd.to_datetime(df['Dt_Customer'])).dt.days/365).round(1)
```

Figure 11: Creating column Tenure from date column

```
df['Category_Breadth'].value_counts().sort_index()
```

```
Category_Breadth
2      12
3     157
4     137
5     485
6    1449
Name: count, dtype: int64
```

Figure 12: Creating Category Breadth column

```
df[['Year_Birth','Age']].head()
```

	Year_Birth	Age
0	1961	65
1	1985	41
2	1975	51
3	1947	79
4	1971	55

Figure 13: Created Age column from Year Birth

Exploratory Data Analysis (EDA):

Income Distribution:

Most customers fall in the income range 30k-80k with a very few high income outliers. The income distribution is slightly right skewed, indicating a small segment of high income customers.

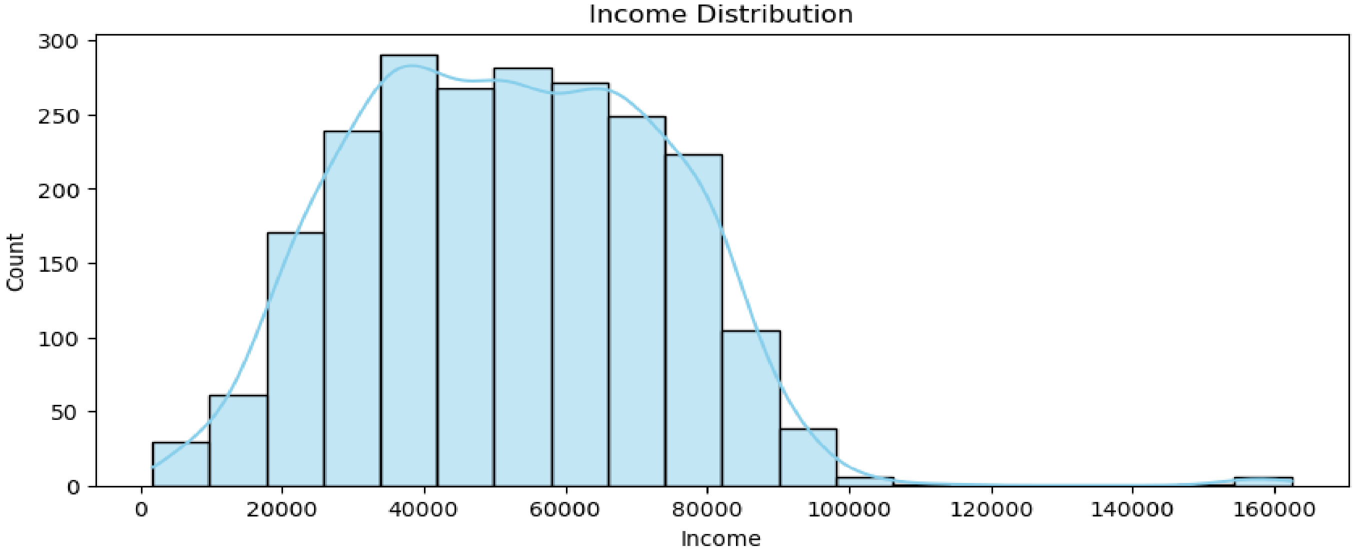


Figure 14: Distribution of customer income

Age Distribution:

Customer age is mostly concentrated between 40 and 70 years, with a peak around 50-55 years. Very few customers are below 35 or above 80 years.

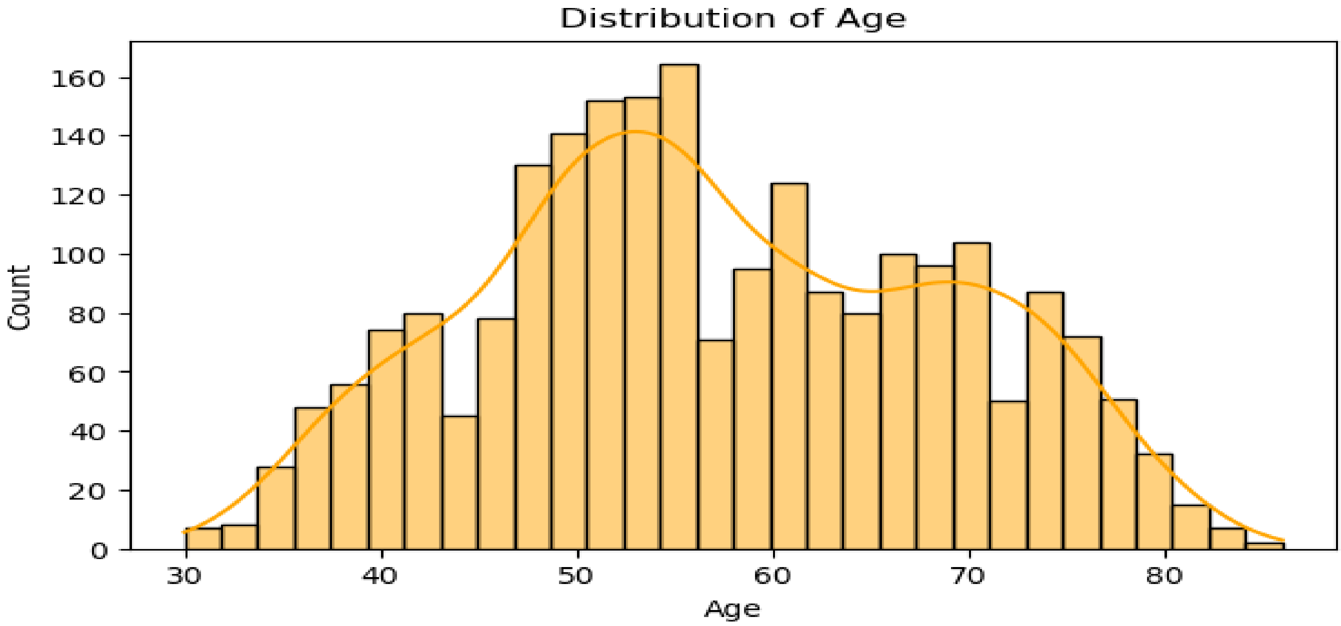


Figure 15: Distribution of customer age

Income vs Total Spend:

Total spend increases with income, showing a strong positive relationship, with most customers are concentrated in range 40k-80k.

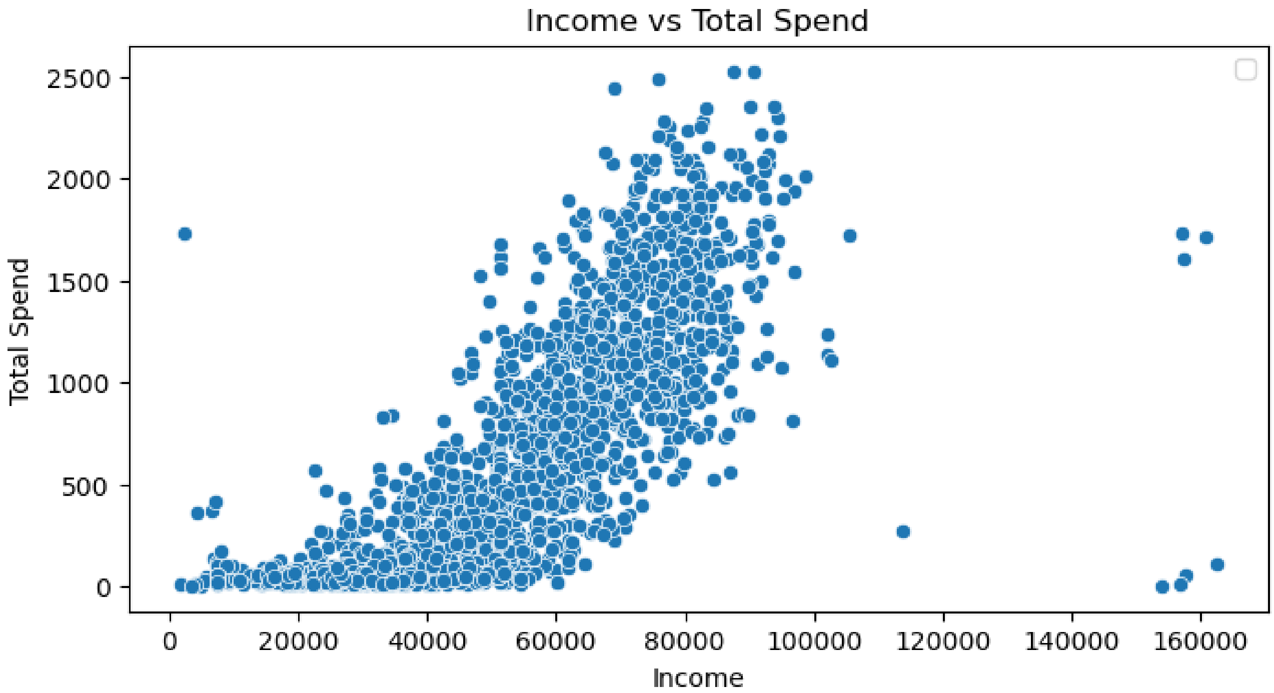


Figure 16: Relationship between income and total customer spend

Education:

Graduation has the highest number of customers with 1127 (50.31%), followed by PhD with 486 (21.70%).

PhD customers show the highest median spending and high response rate of 20.7%, while Graduation and Master have similar median levels. Basic Education has the lowest spending.

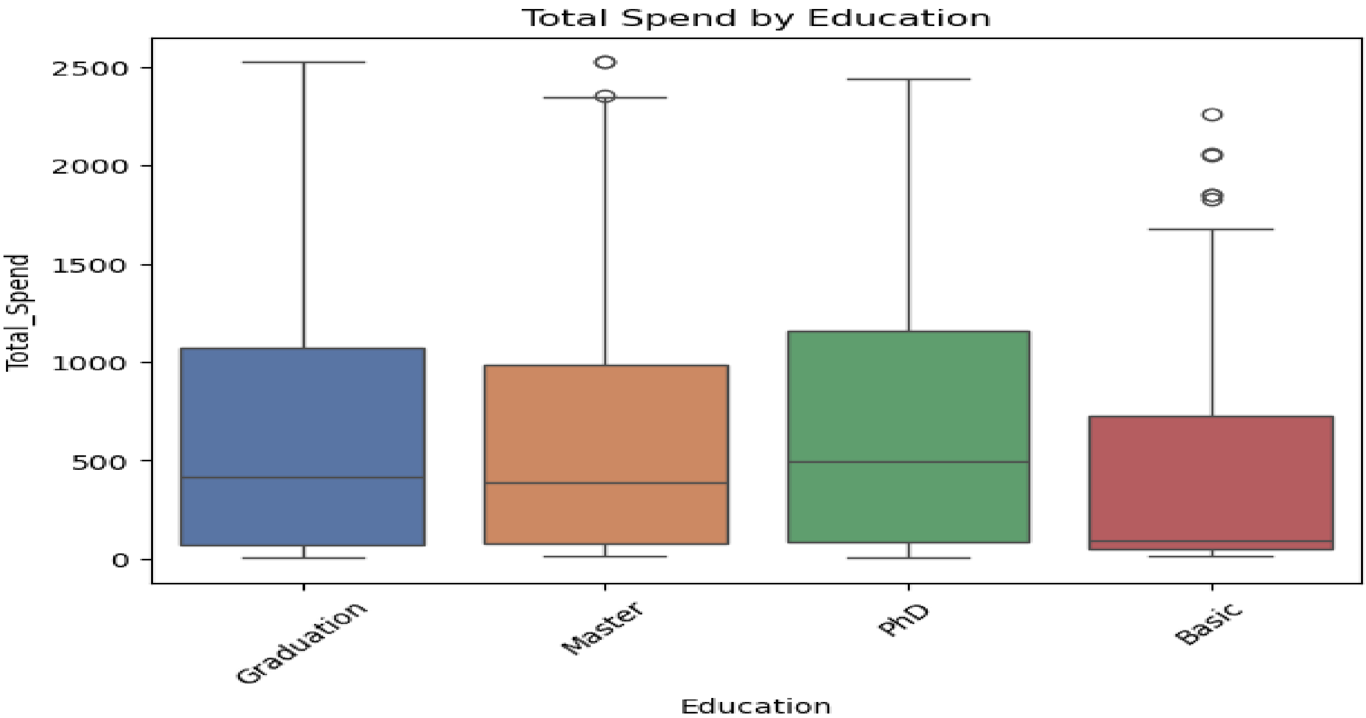


Figure 17: Distribution of customers by education level

Marital Status:

Married customers are the highest with 864 (38.57%), followed by Together with 580 (25.89%). Widow is least with 77 customers.

Widow segment shows highest median spending and high response rate of 24.7%, while other categories such as Single, Married, Together, and Divorced have similar spending levels.

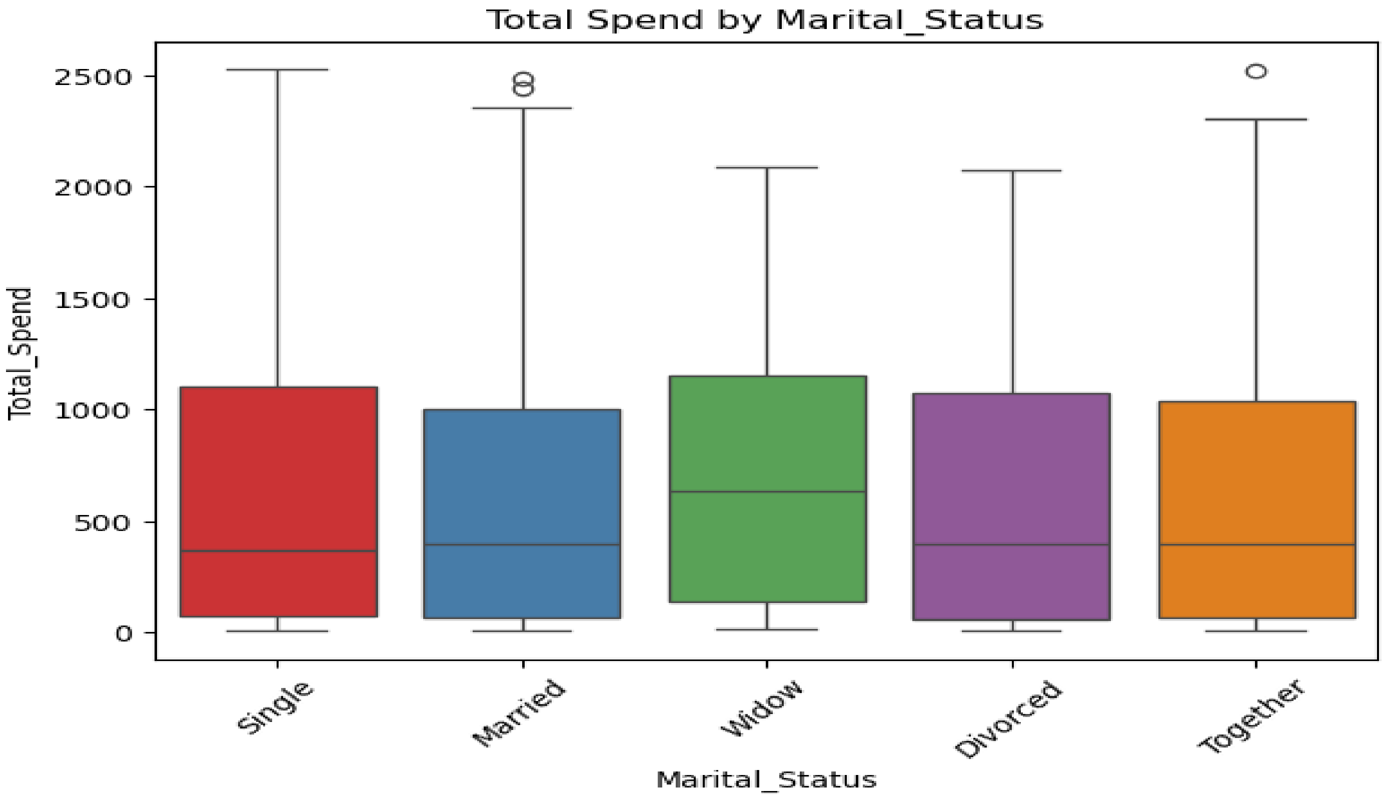


Figure 18: Distribution of customers by Marital Status

Total Spend Distribution:

Wines contribute the highest total spend (0.68M), followed by Meat (0.37M). Other categories like Fish, fruits, Sweets and Gold contribute comparatively lower spending.

Low spending customers are the highest with 1245 (55.58%), followed by Moderate spending customers with 749 (33.44%), while high spending customers are the least with 246 (10.98%).

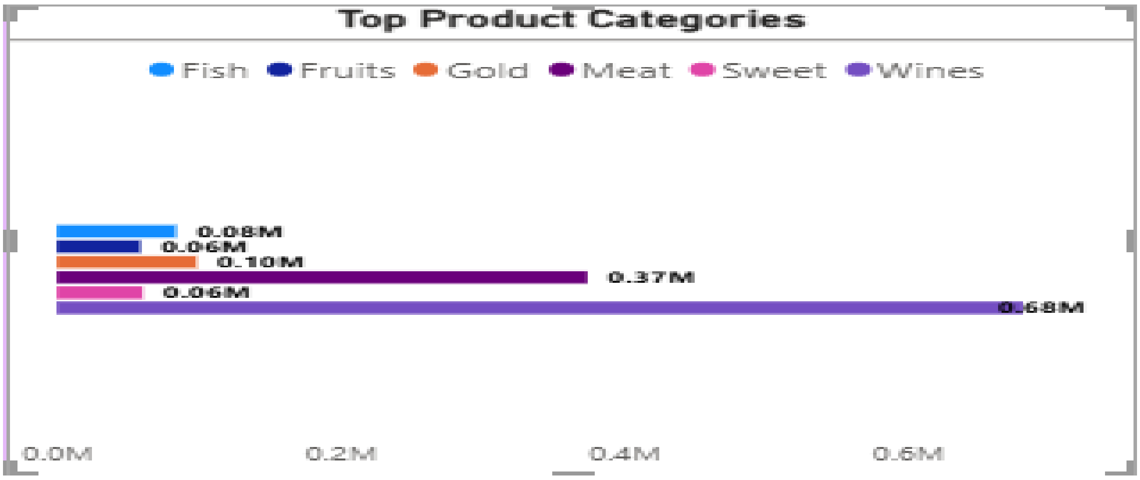


Figure 19: Distribution of total spend among customers

Purchase Channel Usage:

Customers preferred Store purchase (2225), followed by Deal (2194) and Web (2191) purchases. While Catalog usage remains the lowest.

Store channel has the highest total purchase (12970), followed by Web (9150) and Catalog (5963).

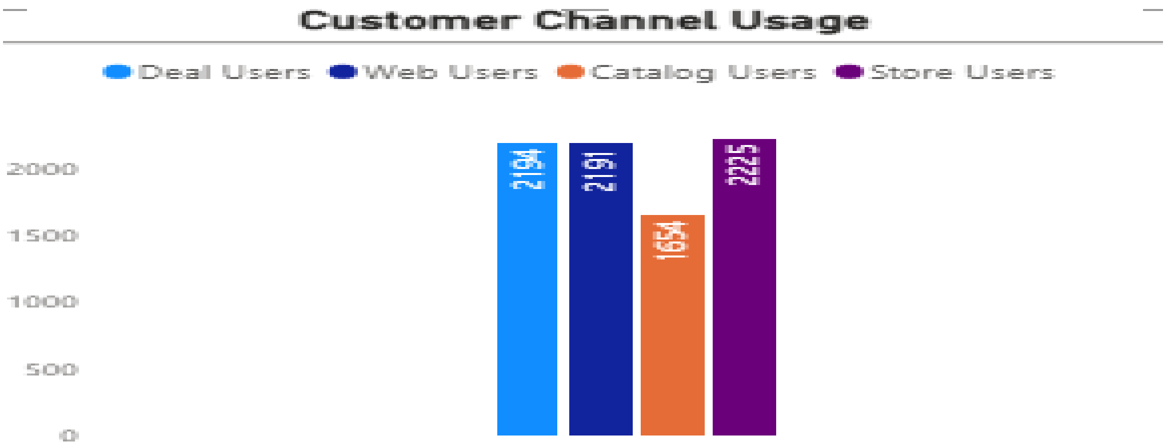


Figure 20: Customer purchase distribution across different channels

Customer distribution by Children:

Most customers have 1 child (1118), making it the largest segment, followed by customer with no children (638).

Customers with no children have the highest average spend, followed by customers with 1 children.

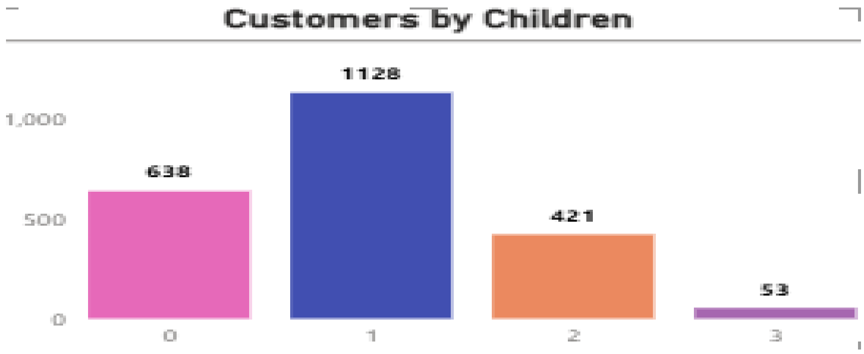


Figure 21: Customer distribution by number of children

<u>Statistical Analysis</u>:

3 Statistical tests were performed:

T- Test Analysis:

1. Total Spend by Children Status
   Result: The result show that customers without children spends more than customers with children. The p-value (p-value<0.05) indicates that there is a significant difference in spending.
   Customers without children spend more than those with children, with an average spend of 1106.02 vs 406.58.
2. Response by Children Status
   Result: The result show a significant difference in response for customers with and without children (p-value<0.05)
   Customers without children have a higher acceptance rate of 26.48% compared to customers with children (10.29%).

```
T-Statistics: -29.126994109752424
P-value: 2.023752068671191e-158
Avg Spend with Kids: 406.57990012484396
Avg Spend without Kids: 1106.0297805642633
```

Figure 22: Total Spend by Children

```
T-Statistics: -9.915238293025915
P-value: 1.0424367047639332e-22
Acceptance rate with Kids: 0.10299625468164794
Acceptance rate without Kids: 0.26489028213166144
```

Figure 23: Response rate by Children

ANOVA (One-Way Analysis):

1. Education vs Total Spend

Result: P-value shows that Education level has a significant impact on spending. Customers with higher education tend to spend more compared to those with basic education.

2. Marital Status vs Total Spend

   Result: P-value is more than 0.05 which shows no significant difference in spending across marital status categories.

```
ANOVA Test Results
F-Statistics: 11.470292045179532
P-value: 1.8361393210251286e-07
```

Figure 24: ANOVA: Education vs Total Spend

```
ANOVA Test Results
F-Statistics: 1.0796029799243507
P-value: 0.3649040489081555
```

Figure 25: ANOVA: Marital Status vs Total Spend

Chi- Square Test Analysis:

1. Education vs Response

   Result: Test shows a significant association between Education and response. Graduation has the highest number of responders i.e. 152, mainly due to its large customer base, while PhD customers show a higher response rate of 20.7% compared to Graduation (13.5%).

2. Marital Status vs Response

   Result: Test indicates a significant relationship between Marital Status and Response. Single customer have the highest number of responders (109), while Widow customers show the highest response rate of 24.7%, despite a smaller group size. Married customers, although the largest segment, have a lower response rate of 11.3%, suggesting that response behavior varies across marital groups.

```
Chi-Square: Education vs Response
chi2: 21.365226601690757
P-value: 8.840050548990529e-05
Degrees of Freedom: 3
```

Figure 26: Chi-Square: Education vs Response

```
Chi-Square: Marital Status vs Response
chi2: 51.51954854585632
P-value: 1.7384196562604576e-10
Degrees of Freedom: 4
```

Figure 27: Chi-Square: Marital Status vs Response

<u>Predictive Modelling</u>:

I. Baseline Logistic Regression Model

Result: The model achieves high accuracy of 85.71% but has very low recall rate of 6.25%, meaning it fails to identify most of the actual buyers. It correctly detects only about 4 out of 64 buyers, missing the majority. Hence, despite good accuracy, the model is not effective for targeting responders.

II. Balanced Logistic Regression Model

Result: This model improves recall rate from 6.25% to 65.62%, meaning it is able to identify a much larger portion of actual buyers. Although the accuracy decreases to 70.31%, the model captures 42 out of 64 buyers, reducing the number of missed customers. This makes the model more effective for marketing purposes.

III. Random Forest Model

Result: This model achieves high accuracy of 85.94% with a moderate recall of 28.12%. It performs better than the baseline Logistic Regression in identifying buyers but still misses a significant number of them. The model correctly identifies around 18 out of 64 buyers, making it more effective than the baseline model but less efficient than balanced Logistic Regression for targeting potential buyers.

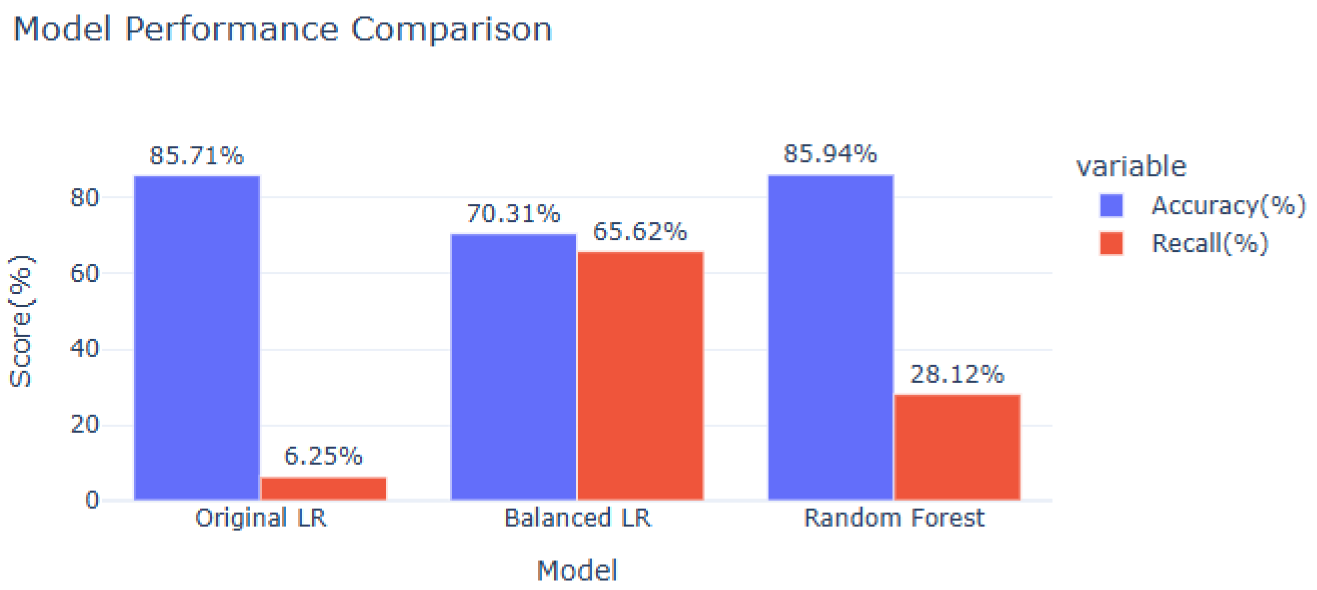


Figure 28: Comparison of the performance of different models

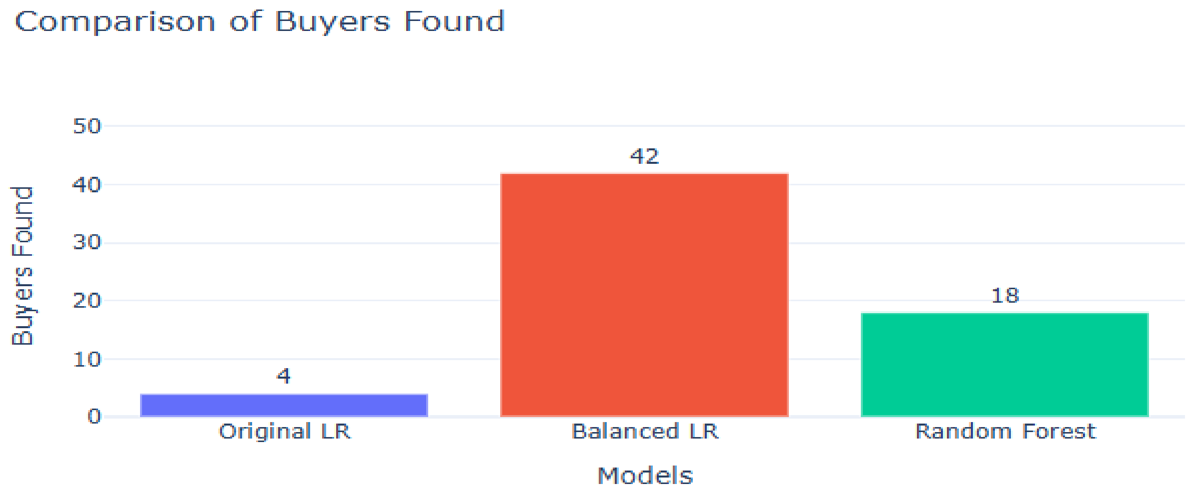


Figure 29: Comparison of the buyers found by different models

Confusion Matrix (Balanced LR):

The model correctly identifies 42 buyers while missing only 22 buyers, showing strong ability to capture potential responders. Although 111 non-buyers are incorrectly classified as buyers, the model prioritizes finding buyers, which is suitable for marketing campaigns.

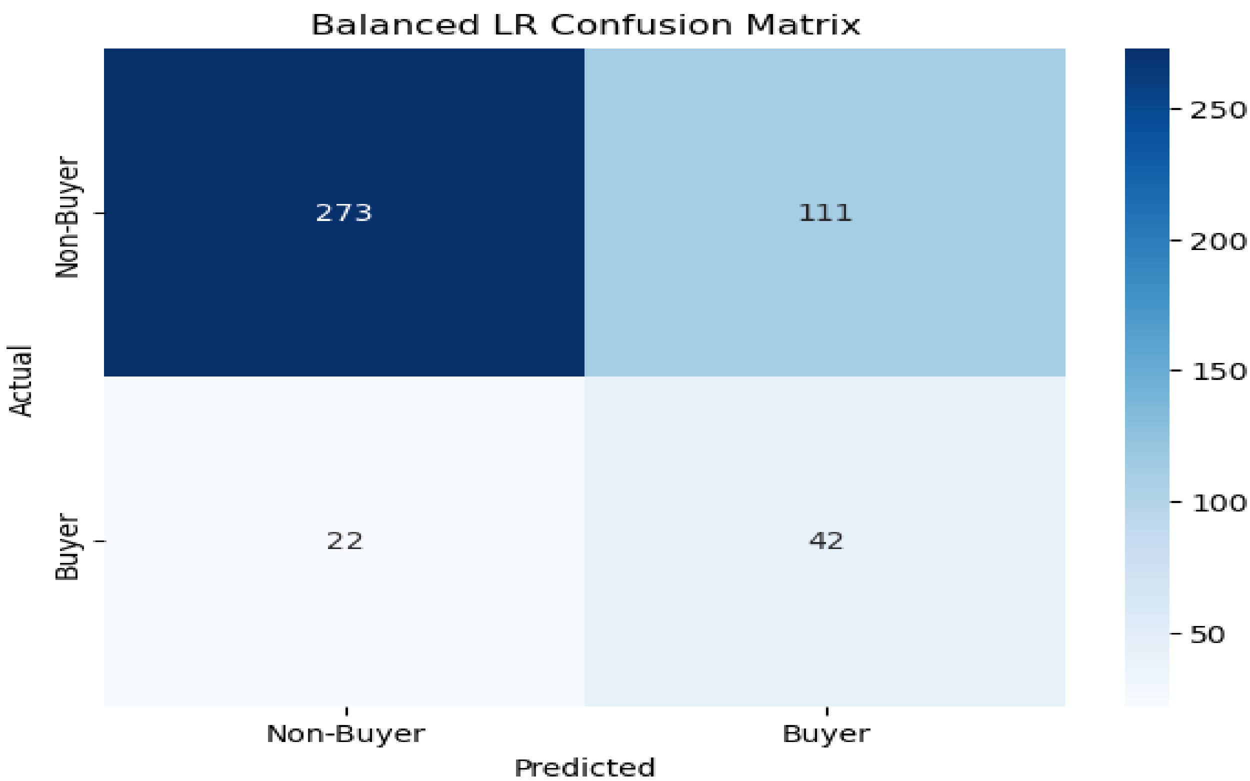
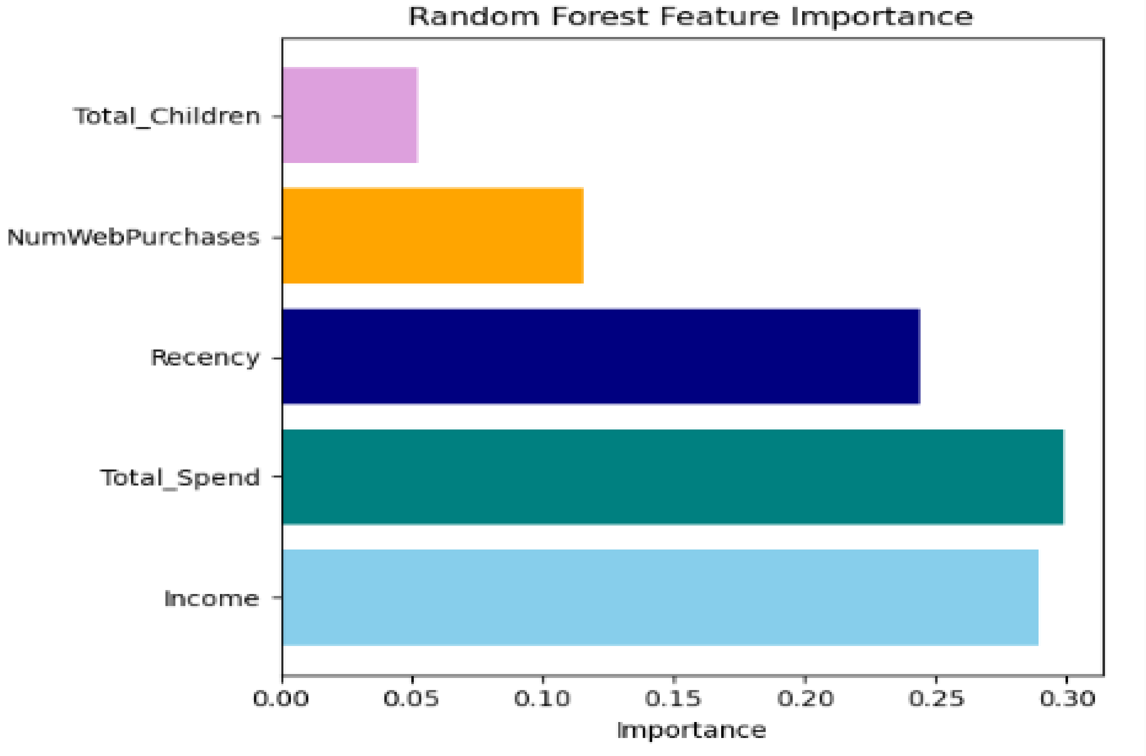


Figure 30: Balanced Logistic Regression Confusion matrix

Key Drivers of Customer Response

Total Spend and Income are the strongest drivers, Recency and Web purchase also plays a positive role, while customers with more children are less likely to respond, as shown by the negative impact.



Feature 31: Showing Random Forest Feature importance

Recommendations:

1. **Income- Based Targeting**
   Provide premium loyalty benefits and personalized campaigns for high-income customers, while offering discounts, EMI options, and flexible payment plans to attract low- income segments.
2. **Re-Engagement Strategy**
   Target inactive and high-recency customers through remainder emails, WhatsApp campaigns, and time-bound offers to bring them back.
3. **Channel Strategy**

Maintain strong in-store presence for trust-based purchases, while promoting digital channels through incentives like free delivery and exclusive online discounts.

4. **Segment-Based Campaigns**
   Focus on Single and Widow customers, as they show higher response rates by offering special value deals and personalized care services.
5. **Model-Based Targeting**
   Adopt Balanced Logistic Regression for campaign targeting, as it identifies more actual buyers, and prioritize recall over accuracy for better customer targeting.

Business Impact

1. The analysis improves business targeting by increasing recall rate from 6.25% to 65.62%, raising identified buyers from 4 to 42, which enables the company to reach more potential customers. By prioritizing recall over accuracy, the model ensures fewer missed opportunities and better campaign effectiveness.
2. To address the low ROI caused by untargeted marketing campaigns, customers were segmented based on income levels, revealing that high-income customers (income > ₹80,000) achieved a response rate of 39.9% compared to 12.2% for low-income customers, enabling more effective audience targeting and improved marketing ROI.
